THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 5: **Arithmetic Progressions** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the AP $\frac{1}{3}, \frac{5}{3}, \frac{9}{3}, \frac{13}{3}, \ldots$, the first term and common difference are: 1

(a) $\frac{1}{3}, \frac{4}{3}$ **(b)** $\frac{1}{3}, \frac{5}{3}$ **(c)** $\frac{4}{3}, \frac{1}{3}$ **(d)** $\frac{5}{3}, \frac{4}{3}$

2. The 18th term of the AP $\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}, \ldots$ is: 1

(a) $35\sqrt{2}$ **(b)** $36\sqrt{2}$ **(c)** $37\sqrt{2}$ **(d)** $34\sqrt{2}$

3. The next term of the AP $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ is: 1

(a) $\sqrt{50}$ **(b)** $\sqrt{40}$ **(c)** $\sqrt{64}$ **(d)** $\sqrt{72}$

4. If $k+2,\ 4k-6,\ 3k-2$ are consecutive terms of an AP, then k is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 5

5. A student states: "The common difference of an AP must always be positive." The correct response is: 1

(a) Correct
(b) Incorrect — it can be positive, negative or zero
(c) Correct for finite APs
(d) Cannot be determined

6. The number of terms in the AP $7, 13, 19, \ldots, 205$ is: 1

(a) 34 **(b)** 33 **(c)** 35 **(d)** 32

7. For the sum of the first 20 natural numbers, Student A gets 210 and Student B gets 200. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. The 4th term from the end of the AP $-11, -8, -5, \ldots, 49$ is: 1

(a) 37 **(b)** 40 **(c)** 43 **(d)** 46

9. If $a_n = 3 + 4n$, the common difference is: 1

(a) 3 **(b)** 4 **(c)** 7 **(d)** 1

10. The sum of the first n even natural numbers is: 1

(a) n^2 **(b)** $n(n+1)$ **(c)** $n(n-1)$ **(d)** $2n^2$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the 31st term of an AP whose 11th term is 38 and 16th term is 73. 2

12. Find the sum of the odd numbers between 0 and 50. 2

13. Which term of the AP $121, 117, 113, \ldots$ is its first negative term? 2

14. A student claims that -150 is a term of the AP $11, 8, 5, 2, \ldots$ Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the sum of all two-digit numbers which are divisible by 7. 3

16. If the sum of the first n terms of an AP is $3n^2 + 5n$, find its first term, common difference and nth term. 3

17. Find the sum: $34 + 32 + 30 + \cdots + 10$. 3

OR

In an AP, $a = 5$, $d = 3$ and $a_n = 50$. Find n and S_n. 3

18. A sum of ₹700 is to be used to give seven cash prizes to students of a school for their overall academic performance. If each prize is ₹20 less than its preceding prize, find the value of each prize. 3

OR

In a school, each section of each class plants as many trees as its class number (Class I section plants 1 tree, Class II section plants 2 trees, …). There are three sections of each class from I to XII. How many trees are planted? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** In a potato race, a bucket is placed at the starting point, 5 m from the first potato; the other potatoes are placed 3 m apart in a straight line. There are ten potatoes. A competitor starts from the bucket, picks the nearest potato, runs back to drop it in the bucket, and continues similarly. 5

(i) Write the distances run for the first three potatoes.

(ii) Find the total distance the competitor runs.

(iii) A student says the answer is 185 m. Find the mistake.

OR

Find the sum of the first 51 terms of an AP whose second and third terms are 14 and 18 respectively. A student says $S_{51} = 51 \times a_{26}$. Verify this claim. 5

20. If the sum of the first 6 terms of an AP is 36 and that of the first 16 terms is 256, find the sum of the first 10 terms. Also find its 20th term. 5

OR

The 17th term of an AP exceeds its 10th term by 7. Find the common difference. If the first term is 3, find the sum of the first 20 terms and check whether 50 is a term of the AP. 5

END OF QUESTION PAPER — Total: 40 Marks